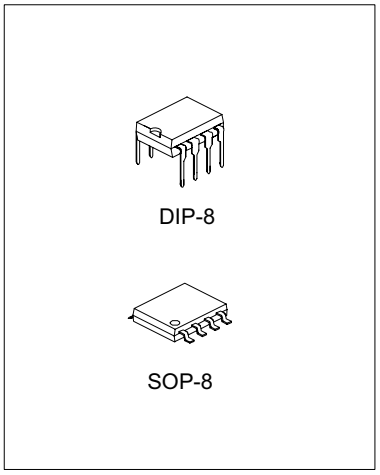
DUAL LOW VOLTAGE POWER AMPLIFIER



DESCRIPTION

The UTC2822H is a monolithic integrated audio amplifier in a 8-pin DIP and SOP package. It is designed for portable cassette players and radios.

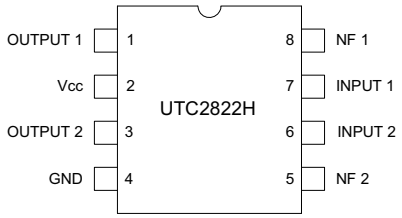
FEATURES

* Wide operating supply voltage:Vcc=1.8V- 6V
* Low crossover distortion
* Low quiescent circuit current
* bridge/stereo configuration

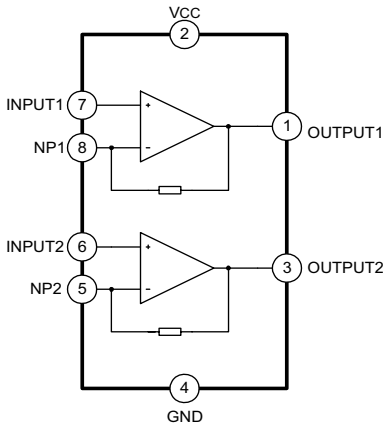
ORDERING INFORMATION

Device	Package
UTC2822	DIP-8-300-2.54
UTC2822E	SOP-8-225-1.27

PIN CONFIGURATION



BLOCK DIAGRAM



ABSOLUTE MAXIMUM RATINGS(Ta=25°C)

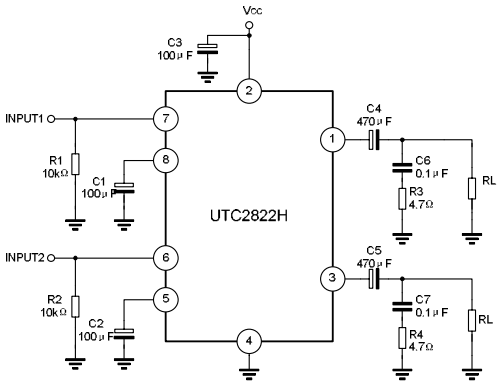
Characteristic	Symbol	Value	Unit
Supply Voltage	V_{CC}	9	V
Output Peak Current	I_O(peak)	1	A
Power Dissipation	P_D	at T_{amb}=50°C 1.0 at T_{case}=50°C 1.4	W
Operating Temperature	Topr	-20~+70	°C
Storage Temperature	Tstg	-40~+150	°C

ELECTRICAL CHARACTERISTICS

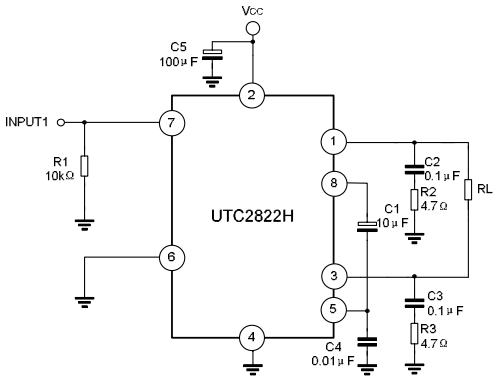
(Ta=25°C,V_{CC}=6V,f=1kHz,unless otherwise specified)

Characteristic	Symbol	Test Conditions	Min	Typ	Max	Units
Operating Supply Voltage	V_{CC}		1.8		6	V
Quiescent Circuit Current	I_{CC}	Vi=0		6.5		mA
Closed Loop Voltage Gain	A_V	Stereo		40		dB
Closed Loop Voltage Gain	A_V	Bridge		40		dB
Channel Balance	CB	Stereo	-1	0	1	dB
Output Power	P_O	Stereo,V_{CC}=3V,R_L=8Ω, THD=10%		56		mW
Output Power	P_O	Stereo,V_{CC}=6V,RL=8Ω, THD=10%		0.35		W
Output Power	P_O	Bridge,V_{CC}=6V,R_L=8Ω, THD=10%		0.7		W
Output Power	P_O	Bridge,V_{CC}=3V,R_L=8Ω, THD=10%		0.15		W
Total Harmonic Distortion	THD	Stereo,R_L=8Ω, Po=0.5W		0.3		%
Total Harmonic Distortion	THD	Bridge,R_L=8Ω, Po=0.5W		0.2		%
Ripple Rejection	RR	Stereo,f=100Hz,C3=100μF	24	30		dB
Total Input Noise Voltage	V_{NI}	BW(-3dB)=20Hz~20kHz		3		μV
Cross Talk	CT	Stereo,f=1kHz		50		dB
Input Resistance	R_i		100			kΩ

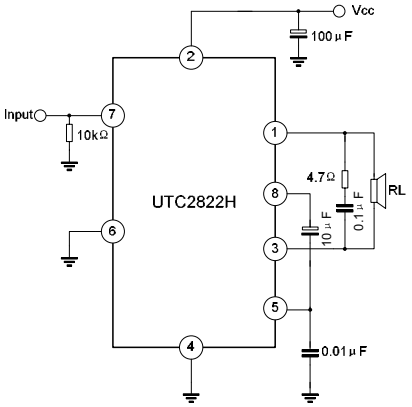
TEST CIRCUIT 1: STEREO



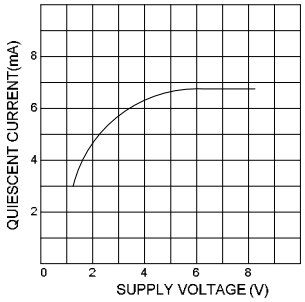
TEST CIRCUIT 2: BRIDGE

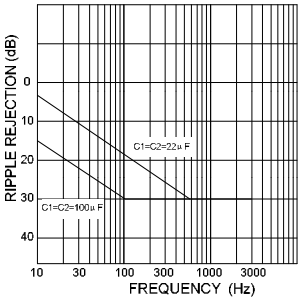


BTL APPLICATION CIRCUIT

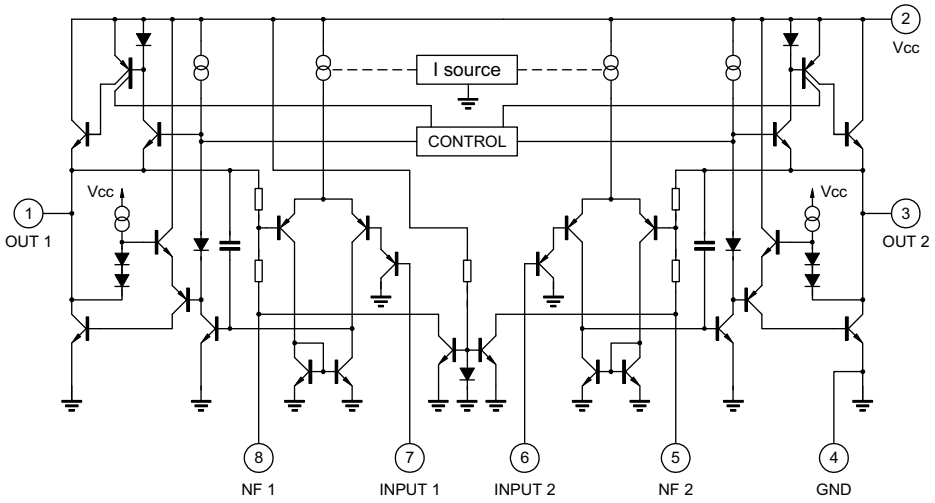


TYPICAL PERFORMANCE CHARACTERISTICS



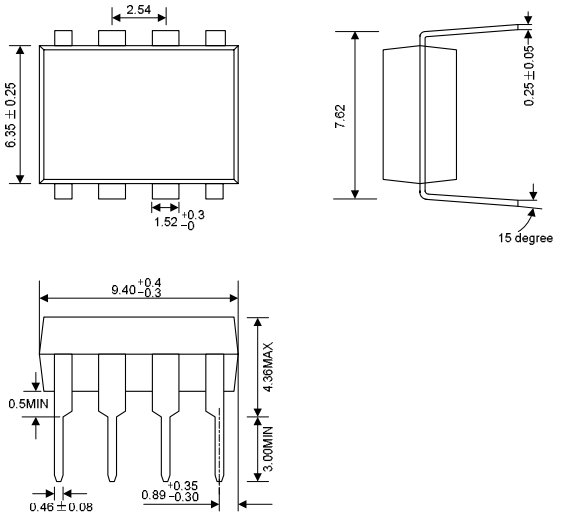


SCHEMATIC DIAGRAM



PACKAGE OUTLINE

DIP-8-300-2.54 UNIT: mm



SOP-8-225-1.27 UNIT: mm

